

Mohamed Gamal Sedeek

AI Engineer | Applied LLM Systems | Senior .NET Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 10+ years of production backend and data engineering, now building applied LLM systems with the same discipline: evaluation set before retriever, confidence intervals on every result, and a reproducibility registry binding each published number to its command, commit and raw output. Shipped a measured Arabic RAG system that benchmarks four retrieval configurations and audits generated citations with a deterministic program rather than an LLM judge. Brings the half of AI engineering that is ordinary engineering: event-driven services at 5,000 messages/sec, CDC pipelines at 10,000 records/min, Kubernetes delivery, and technical ownership of mission-critical government platforms.

CORE EXPERTISE

AI and LLM Engineering: RAG architecture, Retrieval evaluation (Recall@k, MRR, bootstrap confidence intervals), Dense embeddings (multilingual-e5), BM25 and hybrid RRF fusion, Cross-encoder reranking, Citation grounding and abstention design, Schema-constrained output, Local LLM serving (Ollama), sentence-transformers, Hugging Face Transformers, PyTorch (inference), Python, FastAPI, pytest

Backend and Architecture: C#, .NET 6/7/8, ASP.NET Core, REST APIs, Microservices, Clean Architecture, CQRS, DDD, SOLID, MediatR, Entity Framework Core, Dapper

Distributed Systems and Data: Apache Kafka, Debezium CDC, Event-Driven Architecture, SQL Server, PostgreSQL, MongoDB, SSIS/ETL, Data Warehousing, Power BI

Platform and Delivery: Docker, Kubernetes (AKS), Helm, Nginx Ingress, GitHub Actions, GitLab CI, Azure DevOps, CI/CD, Prometheus, Grafana

Security and Quality: JWT, OAuth 2.0, RBAC, xUnit, Moq, Integration Testing, ADRs, Code Reviews, Agile Delivery

APPLIED AI PROJECTS

Design, alternatives and code for all seven projects: mgsm20101.github.io/portfolio/Mohamed-Gamal-Sedeek-Projects-Explained.pdf

Arabic Legal RAG - measured retrieval and citation-forced generation | github.com/mgsm20101/arabic-legal-rag

- Benchmarked four retrieval configurations (BM25, dense multilingual-e5, hybrid RRF, cross-encoder rerank) with 95% bootstrap intervals, then tripled the eval set to 60 questions with 20 held out; at 30 scored questions no ordering survives its interval (Recall@5 0.467-0.767), which overturned the repository's own published headline - and I retracted it in the README.
- Traced the apparent reversal to a composition effect rather than a retrieval change: the colloquial share of scored questions fell from 53% to 30%, and re-weighting the new per-category rates by the old mix put dense back on top and BM25 last.
- Measured a cross-encoder reranker at roughly 55x the median query latency (7,994 ms against 145 ms, n=90; 55-64x across runs) for a quality gain the eval set cannot resolve; shipped dense retrieval.
- Built citation-forced generation (gemma3:4b, greedy, local Ollama) scored by a deterministic audit instead of an LLM judge: 0 fabricated citations in all 21 runs; the failure is omitted citations (11-12 of 30 answers fully grounded). Traced a result that would not reproduce to GPU layer placement in pre-registered experiments: deterministic at each of five placements, while the abstention verdict passed at 2-17 layers and failed at 0 and 34 - so it is reported with its placement.
- Wrote a reproducibility registry binding every published number to its command, commit SHA, environment record and tracked raw result; each harness checks the worktree and the running interpreter itself, and refuses to stamp a result it cannot vouch for.
- Python, sentence-transformers, FastAPI, Ollama, 959 tests under pytest, GitHub Actions.

Intent Router - fine-tuned encoder against a prompted LLM | github.com/mgsm20101/intent-router

- Compared a fine-tuned multilingual DistilBERT against a self-hosted LLM on the same bilingual (Arabic/English) labelled set: the LLM reached 93.8% accuracy with no training, the encoder 71.9% at roughly 130x lower latency per request.
- Served both behind a .NET 8 Minimal API forwarding to a FastAPI service, with schema-constrained JSON output and a code-level validation fallback for invalid labels.

Five smaller measured projects - each public, CI-tested, results at a clean commit | github.com/mgsm20101

- Tool agent over MCP (tool-agent-mcp): guarded loop with allowlist, schema and repeat-call checks; on a local 3B model 4 of 8 tasks completed - 4 of 6 single-step, 0 of 2 multi-step, which the iteration guardrail stopped instead of guessing.
- LLM serving cost (llm-serving-cost): streaming TTFT/throughput benchmark and a cost model reporting capacity as well as break-even; at the measured ~10 tokens/s a local token costs about 4.7x the hosted price. Found and fixed a units bug that put the break-even 1000x too low.
- LLM observability (llm-observability): span-level tracing of a RAG pipeline - generation is 84% of warm latency, and a 42x cold start was the embedding model loading, not the LLM; CI gate on thresholds fixed before the run.
- LoRA on CPU (lora-intent-classifier): LoRA training 0.55% of parameters reached 0.719 accuracy vs 0.750 full fine-tuning, with a 3 MB adapter against 541 MB. .NET 8 assistant (dotnet-ai-assistant): Clean Architecture with the model as a port, 31 model-free tests in CI, 10/10 eval checks against the real model.

PROFESSIONAL EXPERIENCE

Senior .NET Developer | 3i-Vision | Giza, Egypt | Mar 2025 - Present

- Architected an integration layer between WPF desktop and web systems using .NET Core and .NET Framework 4.8, reducing data synchronization errors by 30%.
- Designed secure REST APIs with JWT authentication for high-volume data exchange, supporting 2,000+ concurrent requests across distributed services.
- Optimized SQL Server stored procedures and execution plans, reducing average API response time by 40% on critical operations.
- Applied Clean Architecture, Repository Pattern and Dependency Injection to improve testability and accelerate feature delivery.

Full Stack .NET / Angular Developer | Intlaq - Freelance | Remote | 2022 - 2025

- Delivered Angular 13+ applications against ASP.NET Core APIs with JWT and RBAC for multi-tenant workflows.
- Designed PostgreSQL and EF Core data access for high-traffic endpoints, and automated Docker-based CI/CD with GitHub Actions, taking deployments from hours to minutes.

Senior .NET Developer and Data Engineer | JIC - Judicial Information Center | Cairo, Egypt | 2013 - Feb 2025

- Progressed from junior to senior engineer over 12 years and took technical ownership of mission-critical Ministry of Justice systems.
- Architected REST APIs and microservices for the Ministry of Justice portal with JWT and RBAC, supporting real-time case tracking used daily by 500+ judges and court staff.
- Led delivery of a Complaints Management System on ASP.NET Core and CQRS, automating asynchronous workflows and reducing manual processing by 45%.
- Led a zero-downtime migration from a WebForms monolith to .NET Core microservices running in Linux Docker containers.
- Built SSIS ETL pipelines and a centralized data warehouse, reducing manual reporting effort by 50%, with executive dashboards in Power BI and OBIEE.
- Mentored 5+ junior developers through pairing and pull-request reviews; led Agile delivery and maintained architecture decision records across concurrent projects.

SELECTED TECHNICAL ACHIEVEMENTS

- Designed Kafka-based event-driven microservices in .NET 6 processing 5,000 messages/sec, with containerized horizontal scaling under Kubernetes.
- Implemented a real-time PostgreSQL-to-MongoDB CDC pipeline using Debezium and Kafka, processing 10,000 records/min with no observed data loss.
- Built a distributed ETL solution in .NET Core and Dapper with caching and telemetry-driven query tuning, cutting data-search latency on the reporting workload.
- Deployed a high-availability AKS platform with Helm, Nginx Ingress, Prometheus and Grafana, canary deployments, TLS and Cert-Manager automation.
- Led SQL Server performance work across enterprise systems, reducing the slow-query backlog by 60% and average API response time by 40%.

EDUCATION AND PROFESSIONAL DEVELOPMENT

- Bachelor of Commerce, Cairo University, Faculty of Commerce
- Advanced Data Analysis Nanodegree - Udacity
- Data Analysis Professional Nanodegree - Udacity
- C# Certificate of Completion; Angular, TypeScript, RxJS, WPF and .NET Framework 4.x

LANGUAGES

Arabic: Native | English: Professional Working Proficiency